A FINITE FREE COVER OF THE PUSHFORWARD OF A ONE STEP DEVISSAGE WITH A STRICTLY SMALLER COKERNEL ON THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. The same-chapter referenced definition is located at its \label line inside the pinned chapter file; the cross-chapter Nakayama lemma is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- One step devissage of a finite type module over a point. Same chapter, Definition environment labelled definition-one-step-devissage, source line 780 (Stacks tag 05H4). Supplies the data (Z, Y, i, pi, G) and the affineness of S and Y that the counted proof starts from, plus finite generation of pi_*G; nothing of the construction of alpha is taken from it.
- Nakayama's lemma. Algebra, Lemma labelled algebra-lemma-NAK (canonical Stacks tag 00DV); invoked at proof line 1178. Used once, to promote the vanishing N' tensor kappa(p) = 0 to the vanishing of the localisation N'_p; the choice of r, of the basis x_1, ..., x_r and the dimension comparison are inside the counted span.

Source reference. Original definition 05H4: sources/stacks-flat/pinned-original-source.tex, source lines 780–780. Same-chapter pointer to the definition named in the statement at line 1140; it fixes the hypothesis that S and Y are affine and that pi is finite, all of which the counted proof uses explicitly.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Cross-chapter standard prerequisite invoked once at proof line 1178; it supplies only the promotion of N' tensor kappa(p) = 0 to N'_p = 0, while the construction of alpha and the geometric-irreducibility dimension comparison are retained in the counted proof.

Lemma 0.1. *Let S, X, $\mathcal{F}$, s be as in Definition 05H4. Let $(Z, Y, i, \pi, \mathcal{G})$ be a one step dévissage of $\mathcal{F}/X/S$ over s. Let $\xi \in Y_s$ be the (unique) generic point. Then there exists an integer $r > 0$ and an $\mathcal{O}_Y$-module map*

$$\alpha : \mathcal{O}_Y^{\oplus r} \longrightarrow \pi_*\mathcal{G}$$

such that

$$\alpha : \kappa(\xi)^{\oplus r} \longrightarrow (\pi_*\mathcal{G})_\xi \otimes_{\mathcal{O}_{Y,\xi}} \kappa(\xi)$$

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is an isomorphism. Moreover, in this case we have

$$\dim(\mathit{Supp}(\mathrm{Coker}(\alpha)_s)) < \dim(\mathit{Supp}(\mathcal{F}_s)).$$

Proof. By assumption the schemes S and Y are affine. Write $S = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. As π is finite the $\mathcal{O}_Y$-module $\pi_*\mathcal{G}$ is a finite type quasi-coherent $\mathcal{O}_Y$-module. Hence $\pi_*\mathcal{G} = \widetilde{N}$ for some finite B-module N. Let $\mathfrak{p} \subset B$ be the prime ideal corresponding to ξ. To obtain α set $r = \dim_{\kappa(\mathfrak{p})} N \otimes_B \kappa(\mathfrak{p})$ and pick $x_1, \ldots, x_r \in N$ which form a basis of $N \otimes_B \kappa(\mathfrak{p})$. Take $\alpha : B^{\oplus r} \to N$ to be the map given by the formula $\alpha(b_1, \ldots, b_r) = \sum b_i x_i$. It is clear that $\alpha : \kappa(\mathfrak{p})^{\oplus r} \to N \otimes_B \kappa(\mathfrak{p})$ is an isomorphism as desired. Finally, suppose α is any map with this property. Then $N' = \mathrm{Coker}(\alpha)$ is a finite B-module such that $N' \otimes \kappa(\mathfrak{p}) = 0$. By Nakayama's lemma (Algebra, Lemma 00DV) we see that $N'_{\mathfrak{p}} = 0$. Since the fibre Y_s is geometrically irreducible of dimension n with generic point ξ and since we have just seen that ξ is not in the support of $\mathrm{Coker}(\alpha)$ the last assertion of the lemma holds. $\square$